TROUBLE SHOOTING

Certain abnormal conditions which sometimes interfere with satisfactory engine operation, together with methods of determining the cause of such conditions, are covered on the following pages.

Satisfactory engine operation depends primarily upon:

1. An adequate supply of air compressed to a sufficiently high compression pressure.

2. The injection of the proper amount of fuel at the right time.

Lack of power, uneven running, excessive vibration, stalling at idle speed, and hard starting may be caused by either low compression, faulty fuel injection in one or more cylinders, or lack of sufficient air.

Since proper compression, fuel injection and the proper amount of air are important to good engine performance, detailed procedures for their investigation are given as follows:

Locating a Misfiring Cylinder

1. Start the engine and run it at part load until it reaches normal operating temperature.

2. Stop the engine and remove the valve rocker covers. Discard the gaskets.

3. Check the valve clearance. The clearance should be .009" (two valve cylinder head) or .014" (four valve cylinder head).

4. Start the engine and hold an injector follower down with a screw driver to prevent operation of the injector. If the cylinder has been misfiring, there will be no noticeable difference in the sound and operation of the engine. If the cylinder has been firing properly, there will be a noticeable difference in the sound and operation when the injector follower is held down. This is similar to short-circuiting a spark plug in a gasoline engine.

5. If the cylinder is firing properly, repeat the procedure on the other cylinders until the faulty one has been located.

6. Provided that the injector operating mechanism of the faulty cylinder is functioning satisfactorily, remove the fuel injector and install a new one by performing the removal and installation procedure outlined under *Fuel System.*

7. If installation of a new injector does not eliminate misfiring, check the compression pressure.

Checking Compression Pressure

Compression pressure is affected by altitude as shown in Table 1.

Check the compression pressure as follows:

1. Start the engine and run it at approximately one-half rated load until normal operating temperature is reached.

Minimum Compression Pressure (psi) at 600 rpm		Altitude, Feet Above Sea Level
Standard and Turbocharged Engines	"N" Engines	
425	515	0
395	480	2,500
365	440	5,000
340	410	7,500
315	380	10,000

TABLE 1

Cylinder	Gage Reading
1L	445 psi
1R	440 psi
2L	405 psi
2R	435 psi
3L	450 psi
3R	445 psi

TABLE 2

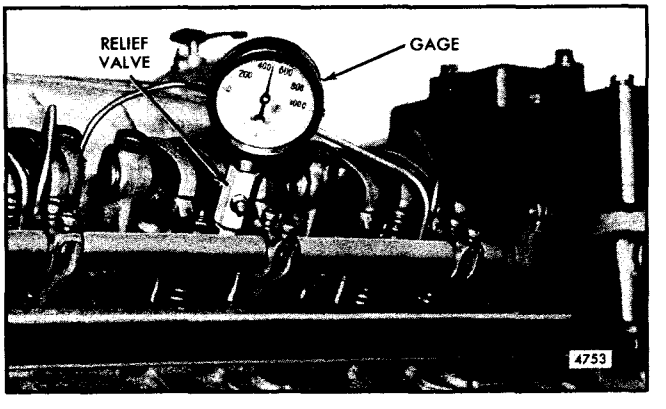


Fig. 1 - Checking Compression Pressure